

Multiple Choice (10 marks)

1. As the number of training examples goes to infinity, your model trained on that data will have:
 - (a) Lower variance
 - (b) Higher variance
 - (c) Same variance
 - (d) None of the above
2. In Yann LeCun's cake, the cherry on top is
 - (a) reinforcement learning
 - (b) self-supervised learning
 - (c) unsupervised learning
 - (d) supervised learning
3. Let us say that we have computed the gradient of our cost function and stored it in a vector g . What is the cost of one gradient descent update given the gradient?
 - (a) $O(D)$
 - (b) $O(N)$
 - (c) $O(ND)$
 - (d) $O(ND^2)$
4. MLE estimates are often undesirable because
 - (a) they are biased
 - (b) they have high variance
 - (c) they are not consistent estimators
 - (d) None of the above
5. Statement 1| RELUs are not monotonic, but sigmoids are monotonic. Statement 2| Neural networks trained with gradient descent with high probability converge to the global optimum.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True

True / False (10 marks)

Answer True or False

6. True or False: DenseNets generally require more memory than ResNets due to their high connectivity patterns.
7. True or False: The numerical output of a sigmoid node in a neural network is unbounded, encompassing all real numbers.
8. True or False: Batch Normalization was used in the original ResNet paper, rather than Layer Normalization.
9. True or False: Predicting the amount of rainfall in a region based on various cues is a supervised learning problem.
10. True or False: The convergence of a neural network is significantly influenced by the choice of learning rate during training.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the validity of the ID 3 algorithm in finding optimal decision trees and analyze the implications of continuous probability distributions with nonzero density functions.
12. Discuss how you would modify the EM algorithm to compute MAP estimates instead of maximum likelihood estimates, focusing on the necessary changes in the maximization step.

End of Paper